

Adult Income Data Cleaning and EDA

Exploratory Data Analysis Report

UCI Adult Income Dataset

1. Project Overview

This project uses the UCI Adult Income dataset to perform data collection, data cleaning, validation, feature engineering, and exploratory data analysis.

The final analysis dataset contains 48,842 records and 20 columns.

The main question is: Which demographic, education, and work characteristics are associated with an annual income above USD 50K?

2. Data Cleaning

The official adult.data and adult.test files were combined into a single analysis dataset while preserving the source split for traceability.

Whitespace was removed from categorical values and the UCI missing-value marker '?' was converted to missing values.

Missing workclass, occupation, and native_country values were replaced with 'Unknown' so that the affected records could be retained.

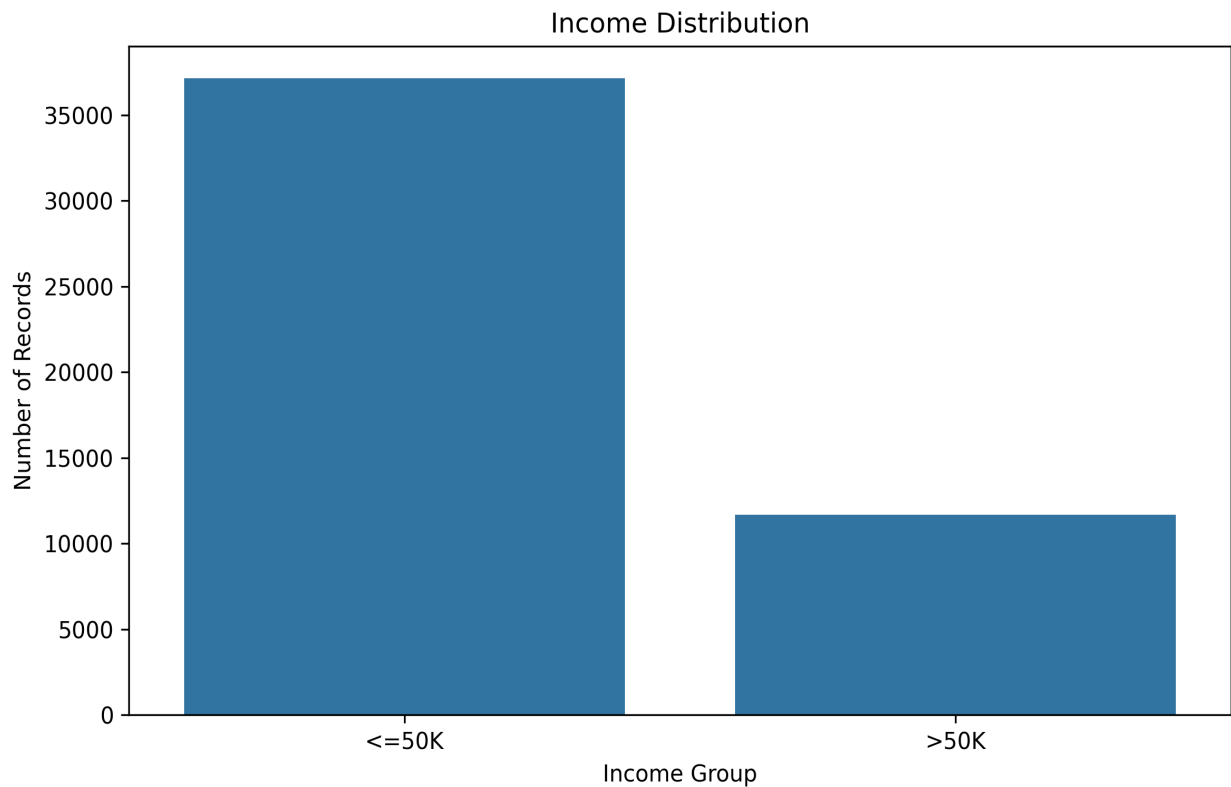
The final cleaned dataset contains 0 remaining missing values.

29 exact duplicate rows were identified and retained because the dataset does not contain a unique person identifier.

3. Overall Income Distribution

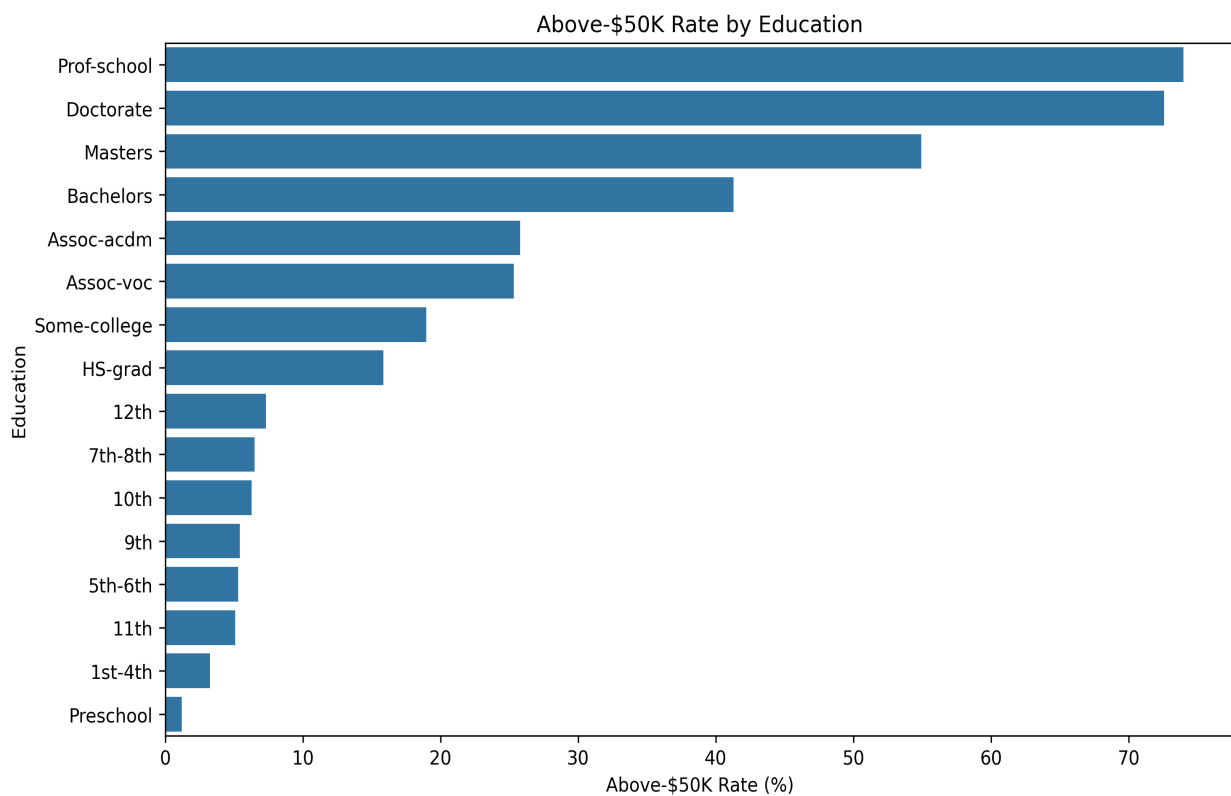
The dataset contains 48,842 records. 11,687 records have income above USD 50K, while 37,155 records have income at or below USD 50K.

The above-USD-50K rate is 23.93%.



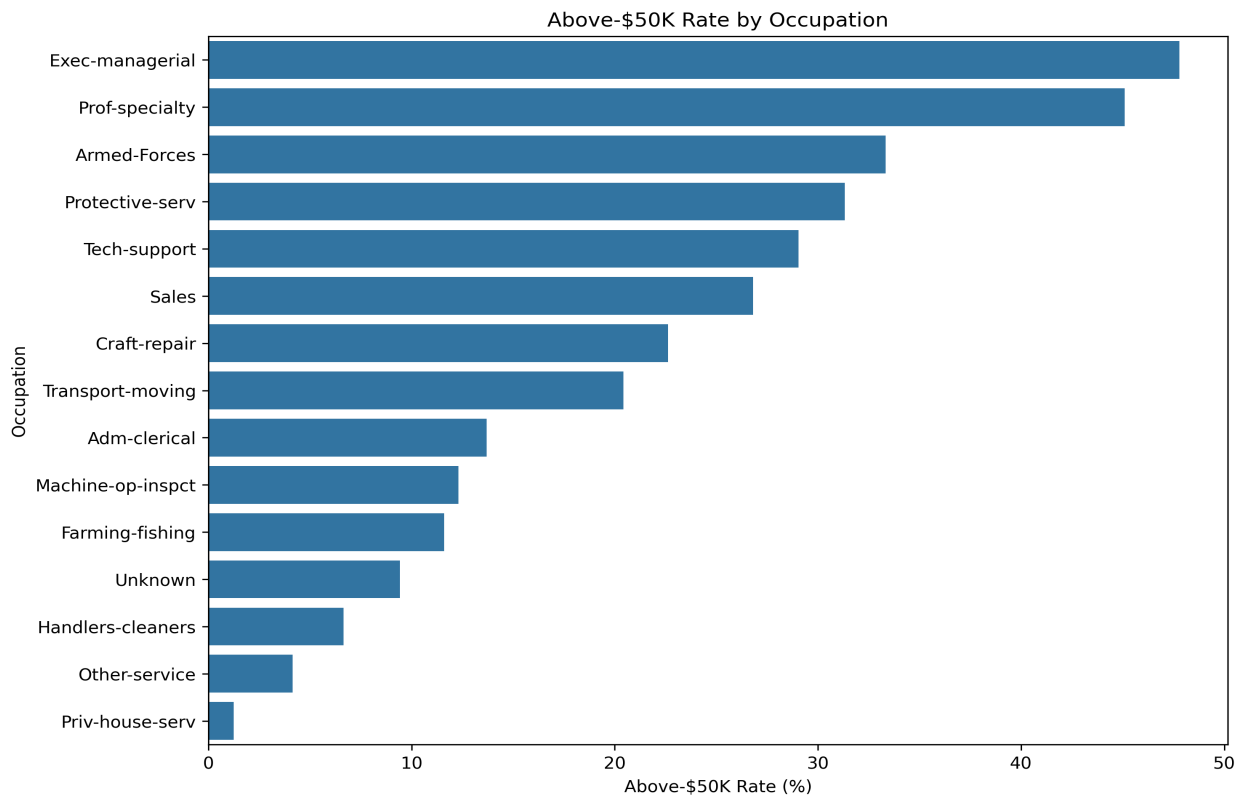
4. Education and Income

The above-USD-50K rate varies across education categories. The education summary table was calculated using the proportion of records with the high_income indicator equal to 1.



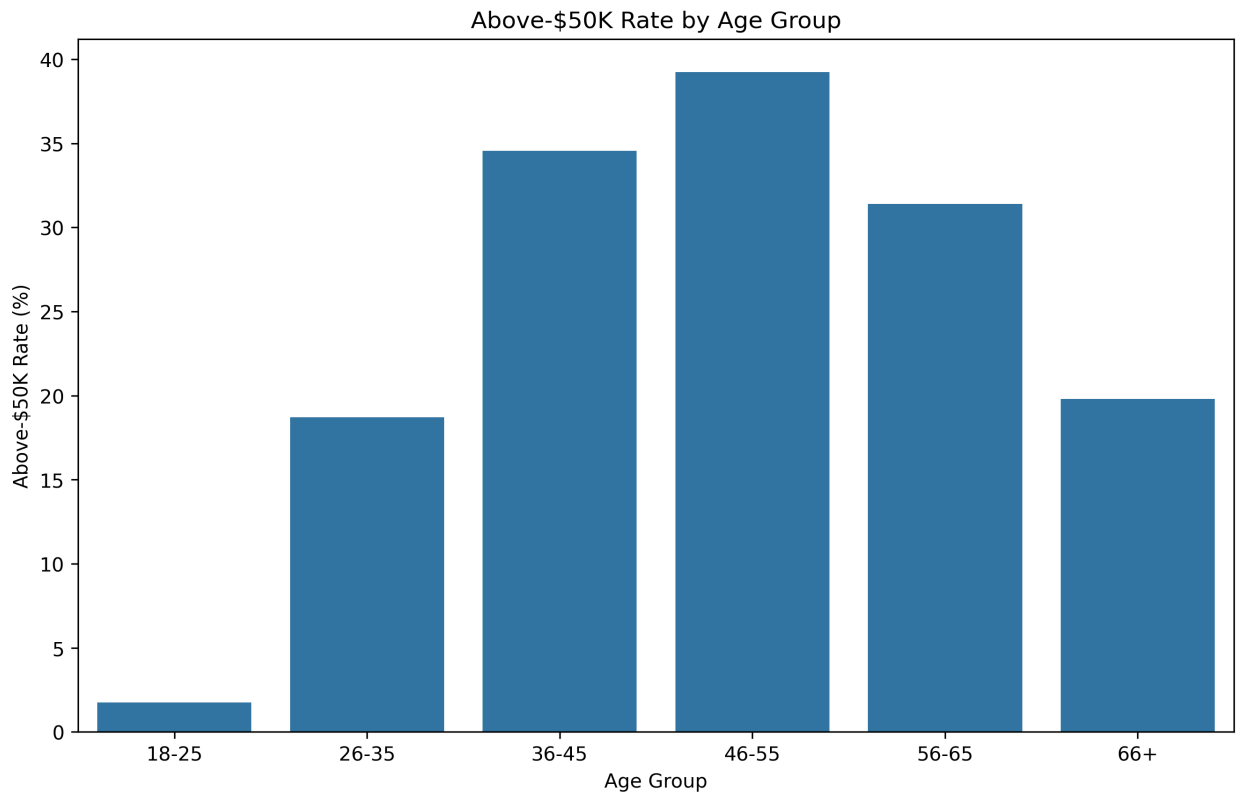
5. Occupation and Income

The above-USD-50K rate varies across occupation categories. The analysis considers both the number of records and the proportion above USD 50K.



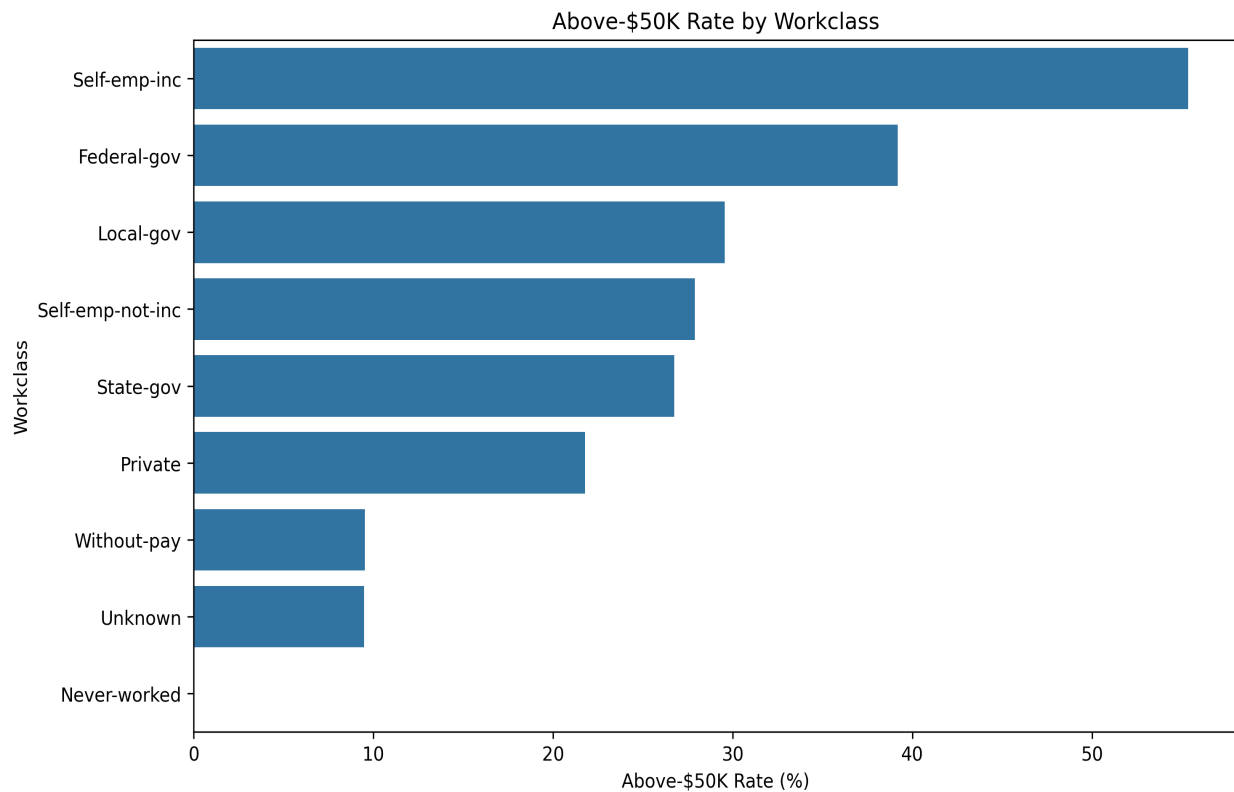
6. Age Group and Income

The above-USD-50K rate varies across the defined age groups. These differences describe patterns observed within the dataset.



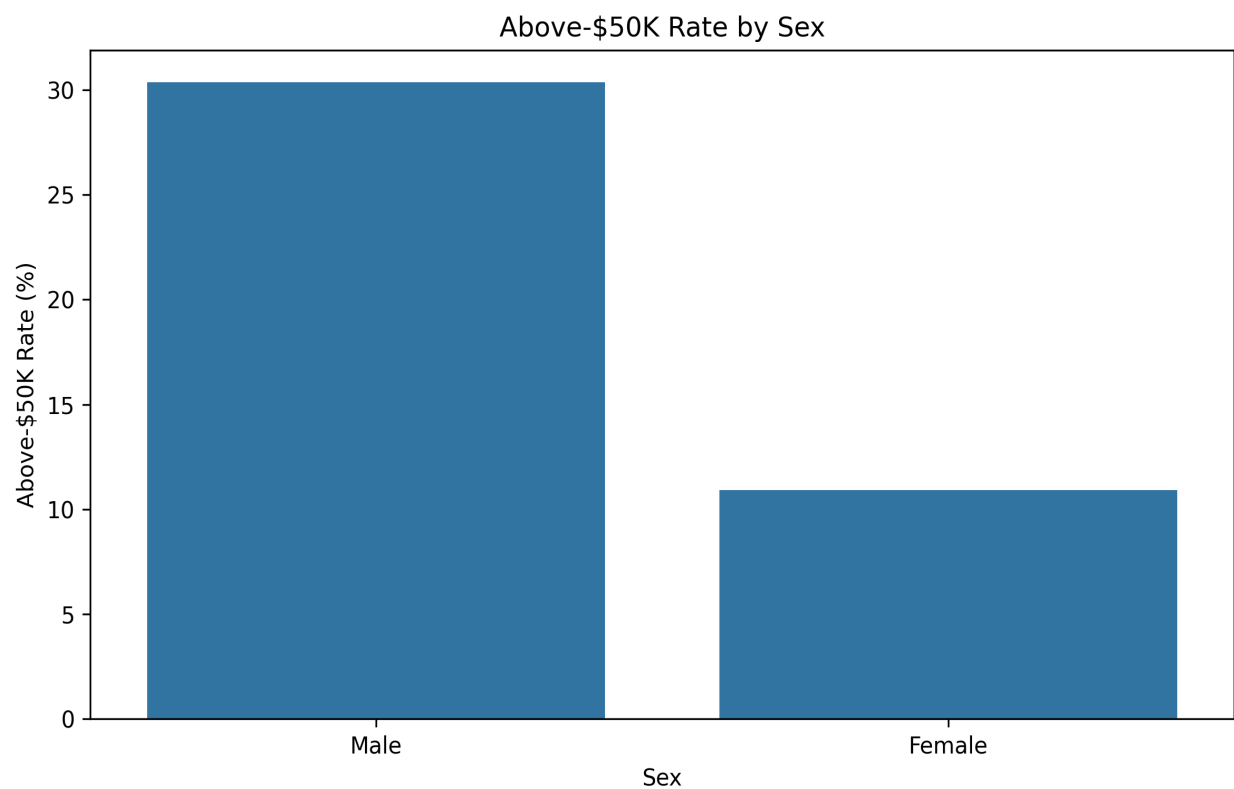
7. Workclass and Income

The above-USD-50K rate differs across workclass categories, including private, government, self-employment, and other workclass categories.



8. Sex and Income

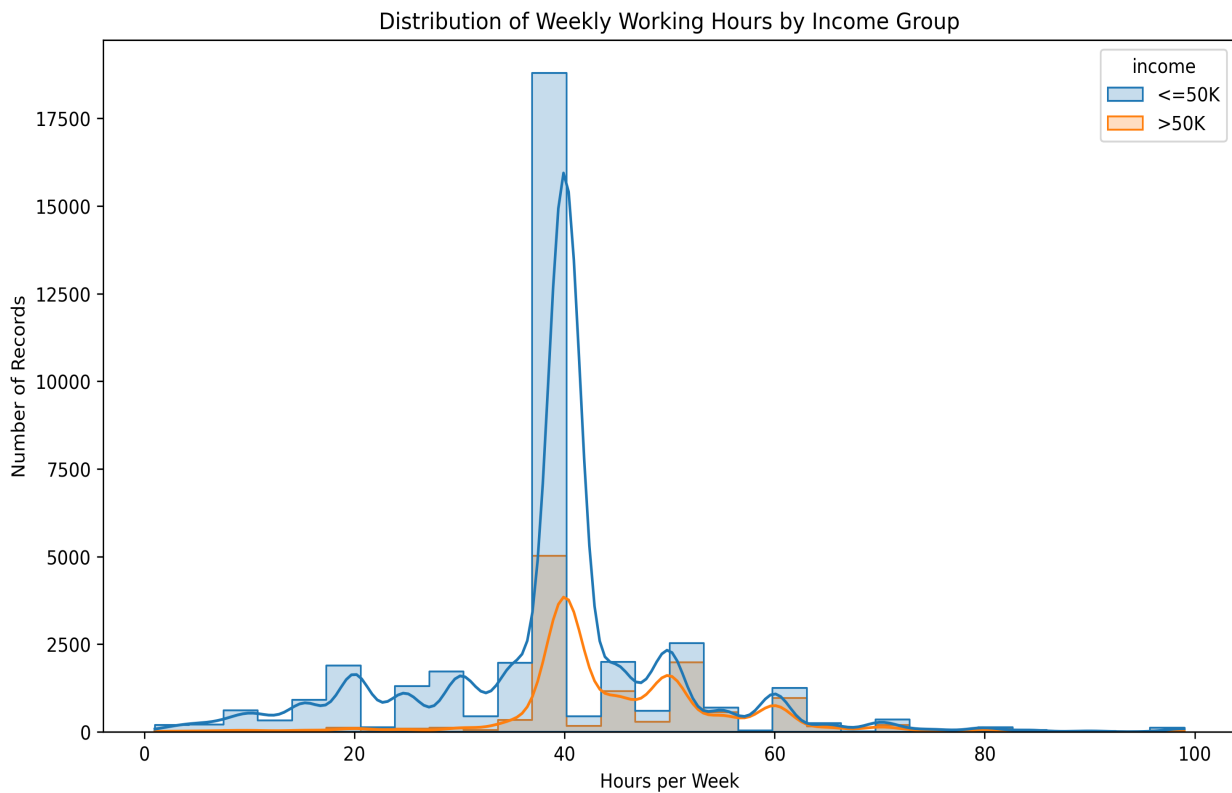
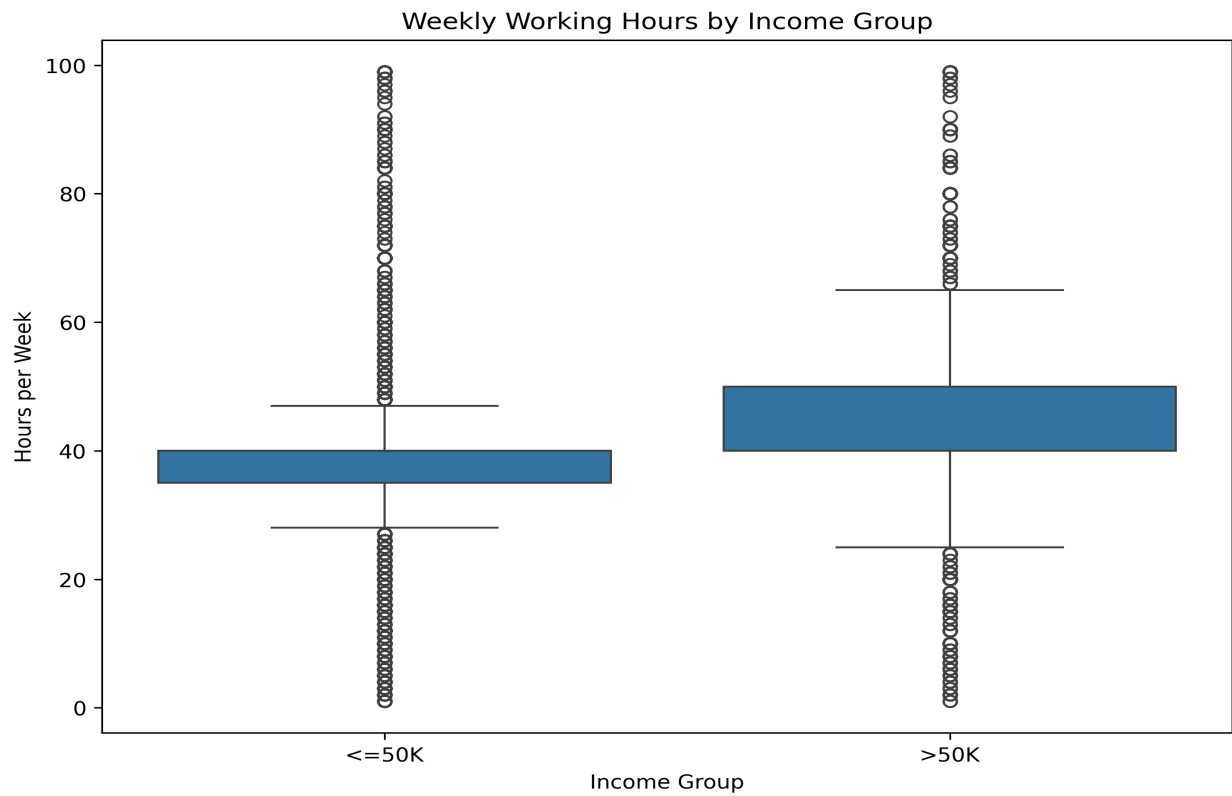
The calculated above-USD-50K rate is 30.38% for male records and 10.93% for female records in this dataset.



9. Weekly Working Hours

The mean weekly working time is 38.84 hours for the $\leq 50K$ group and 45.45 hours for the $>50K$ group.

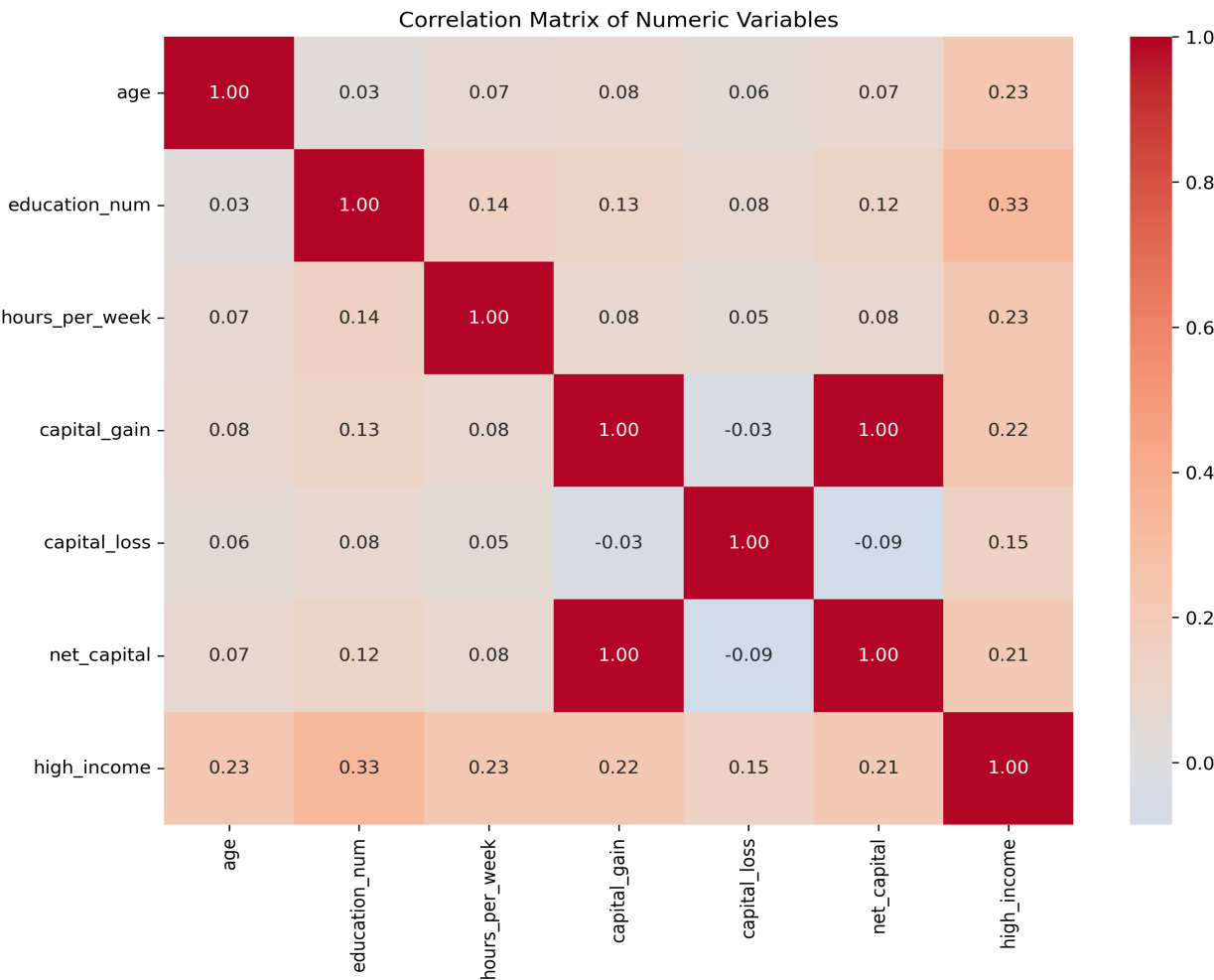
The median is 40 hours for the $\leq 50K$ group and 40 hours for the $>50K$ group.



10. Numeric Relationships

Correlation analysis was performed for age, education_num, hours_per_week, capital_gain, capital_loss, net_capital, and high_income.

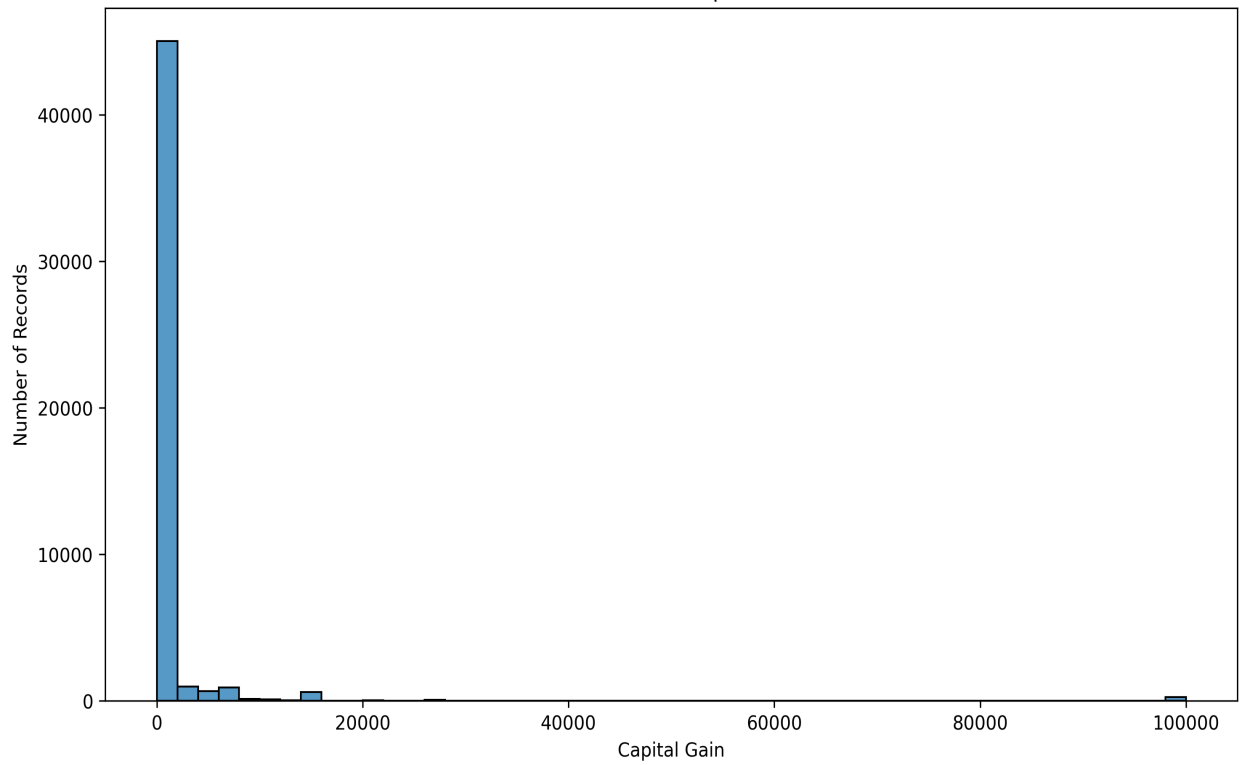
The correlations with high_income were approximately 0.23 for age, 0.33 for education_num, 0.23 for hours_per_week, 0.22 for capital_gain, 0.15 for capital_loss, and 0.21 for net_capital.



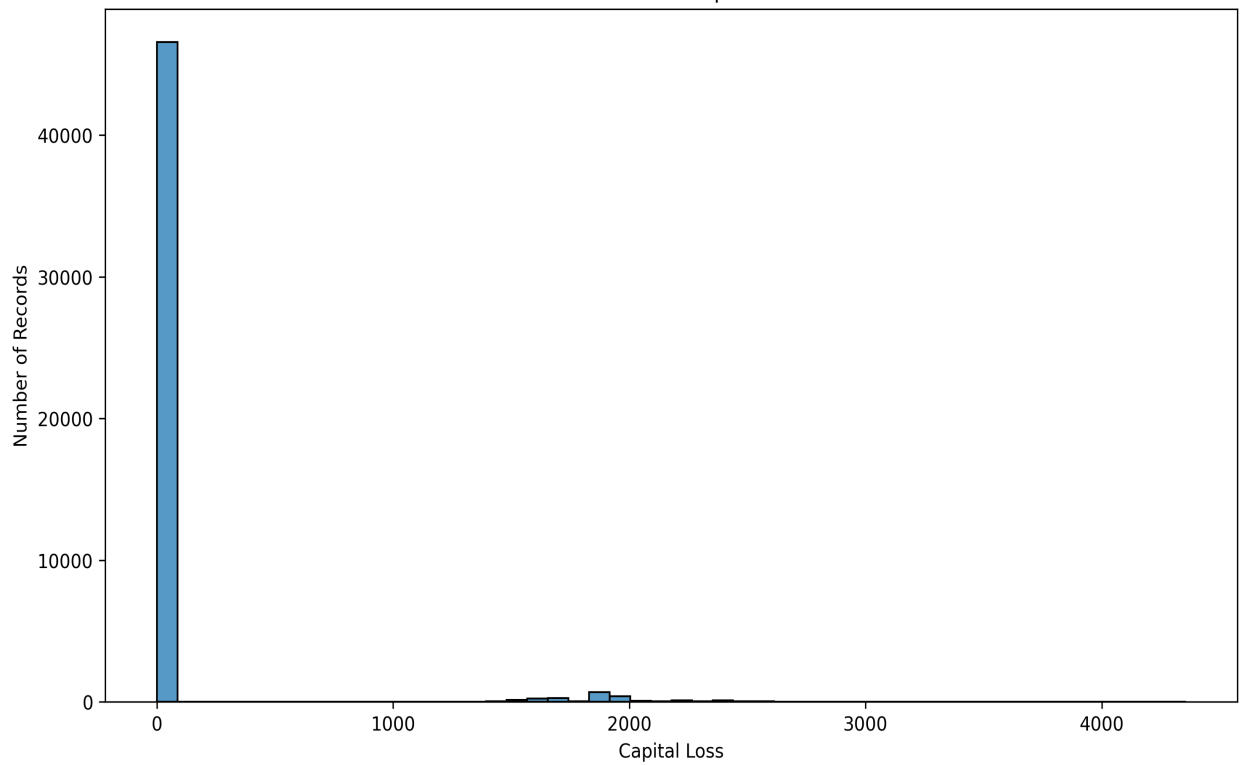
11. Capital Gain and Capital Loss

Capital gain and capital loss distributions are strongly concentrated around zero, with a smaller number of records having substantially larger values.

Distribution of Capital Gain



Distribution of Capital Loss



12. Limitations

- The dataset contains historical census information and may not represent current income patterns.
- The analysis is descriptive and identifies associations rather than causal relationships.
- The dataset contains demographic variables that require careful interpretation.
- Missing categorical values were represented as Unknown rather than deleting affected records.
- Exact duplicate rows were retained because there is no unique person identifier.
- The fnlwgt census weight was not applied to the visual summaries.
- Correlation measures linear association and does not establish causation.
- The income target uses the dataset's $\leq 50K$ and $> 50K$ categories.
- This project does not perform predictive modeling or evaluate predictive performance.

13. Conclusion

This project completed an end-to-end data cleaning and exploratory analysis workflow using the UCI Adult Income dataset.

The analysis examined income distribution and compared the above-USD-50K rate across education, occupation, age group, workclass, and sex. Weekly working-hour distributions and numeric relationships were also examined.

The findings describe patterns and associations within this historical dataset. They should not be interpreted as evidence of causal relationships or as representative estimates of the current population.